Interactions between tumor-associated macrophages and tumor cells in glioblastoma: unraveling promising targeted therapies.

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Glioblastoma (GBM) is the deadliest primary malignant central nervous system (CNS) tumor with a median overall survival of 15 months despite a very intensive therapeutic regimen including maximal safe surgery, radiotherapy, and chemotherapy. Therefore, GBM treatment still raises major biological and therapeutic challenges. Areas covered: One of the hallmarks of the GBM is its tumor microenvironment including tumor-associated macrophages (TAM). TAM, accounting for approximately 30% of the GBM bulk cell population, may explain, at least in part, the immunosuppressive features of GBMs. The TAM are active and highly plastic immune cells and include two major ontogenetically different cell populations: (i) microglia and, (ii) monocytes-derived macrophages (MDM). TAM recruited to the tumor bulk can be reprogramed by GBM cells resulting in an ineffective anti-tumor response. Interestingly, interactions between TAM and GBM cells promote tumor oncogenesis (i.e. tumor cells proliferation and migration/invasion). This review aims to explore TAM targeting in GBM as a promising therapeutic option in the near future. Expert Commentary: A better understanding of TAM-GBM interactions and dynamics will certainly uncover new anti-GBM therapeutic avenues.

Juvenile and adult-onset scleroderma: Different clinical phenotypes.

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Systemic sclerosis (SSc) represents extremely rare disease with majority of data coming from adults. Studies comparing juvenile- (jSSc) and adult-onset (aSSc) patients are limited. We aimed to compare clinical features, treatment modalities and survival rates of jSSc and aSSc patients. A retrospective study among pediatric and adult Scl patients has been performed. Demographic characteristics, clinical features, autoantibody profiles, and treatment data were retrieved from the databases. Survival analysis was done using Kaplan-Meier plot and factors associated with mortality were identified with multiple regression analysis. A total of 158 adults and 58 juvenile Scl patients were identified. The mean age at the disease onset was 37±14.7 vs. 8.8 ± 4.1 years, mean age at diagnosis 42±15.2 vs. 10.4 ± 3.8 years and mean follow-up duration was 6.3 ± 4.9 years vs. 6.6 ± 4.9 years for aSSc and jSSc patients, respectively. The frequency of interstitial lung disease (ILD) (50.9% vs 30%, $p<0.001$) and systemic hypertension (17.9% vs 0, $p = 0.009$) was significantly higher among aSSc. While aSSc patients had presented mostly with limited cutaneous subset (74.1%), diffuse cutaneous subset was the dominant subset among jSSc (76.7%), ($p<0.001$). The mortality rate was significantly higher among adults ($p = 0.005$). The ILD ($p = 0.03$) and cardiac insufficiency ($p = 0.05$) were independent risk factors of mortality in both aSSc and jSSc patients. Juvenile and adult-onset Scl represent rarely seen conditions with different clinical phenotypes. Pediatric patients with LS are more commonly seen by pediatric rheumatologists, in contrary to adults. Diffuse disease subset is the dominant form among juvenile patients, whereas limited form is the main disease subset among adults. On the other hand, juvenile-onset patients have a better survival than those with adult-onset.

The clinical characteristics and outcomes of patients with systemic sclerosis with myocardial involvement.

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Myocardial involvement (MI) is the primary cause of death in patients with systemic sclerosis (SSc). We analyzed patients with SSc and MI to identify their characteristics and outcome. We retrospectively collected data from SSc patients with MI admitted to Peking Union Medical College Hospital between January 2012 and May 2021. SSc patients without MI were randomly selected as controls after matching age and gender at a ratio of 1:3. In total, 21 SSc patients (17 females) with MI were enrolled. The mean age at SSc onset was 42.3 ± 15.1 years old. Compared with controls, myositis (42.9% vs. 14.3%, P = 0.014) and elevation of CK (33.3% vs. 4.8%, P = 0.002) were more common in patients with MI. Of the 7 patients without cardiovascular symptoms, 3 /5 showed elevations in cardiac troponin-I (cTnI), 6 showed elevations of N-terminal brain natriuretic peptide (NT-proBNP). Eleven patients were followed up for a median period of 15.5 months and four patients developed newly occurring left ventricular ejection fraction (LVEF) < 50%. One third of SSc patients with MI were asymptomatic. Regular monitoring of CTnI, NT-proBNP and echocardiography is helpful for the diagnosis of MI during the early stages. Its prognosis is poor.

Prevalence and gender - specific analysis of a systemic sclerosis cohort in Latvia.

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Systemic sclerosis (SSc) is considered by many to be one of the most severe autoimmune rheumatic diseases with lower prevalence observed in Northern Europe. No previous studies on the prevalence of SSc in Latvia have been conducted and the aim was to study the demographic and clinical data of patients with SSc in northeastern Europe country. This study was conducted in two main Latvian hospitals for adults and includes patients with SSc who were consulted between 2016 and 2021. During the study period, 159 patients with SSc were consulted. The point prevalence on 1 January 2021 was 84.0 per million. Female to male ratio was 4.67:1, and highest gender ratio was observed in the age group 70-79-year (6.75:1). Antinuclear antibodies were present in 82.58% of patients, without gender difference. Centromere pattern was more frequently observed in females (40.19% vs. 19.04%), in contrast to speckled pattern (50.98% vs. 57.14%). At disease onset females tended to be younger (46.51 ± 13.52) than males (50.5 ± 16.64). Males had more diffuse cutaneous subtype, interstitial lung disease, pulmonary hypertension and esophageal dysmotility. More than half of patients received treatment with glucocorticoids at any point of the disease (68.31%), without gender difference. Systemic sclerosis is less common in Latvia than in other countries and regions. Due to its location, the data from Latvia are consistent with a north-south gradient in Europe. Gender ratio differences persisted in older age groups as well. Antinuclear antibodies presence did not differ between genders, but in female's centromere pattern was much more likely to be present. Males had more severe disease course, but in both genders more than half of patients received treatment with GCs at any point of the disease.

Early- versus late-onset systemic sclerosis: differences in clinical presentation and outcome in 1037 patients.

PMID: 24646463 | Indexed in PubMed

Peak age at onset of systemic sclerosis (SSc) is between 20 and 50 years, although SSc is also described in both young and elderly patients. We conducted the present study to determine if age at disease onset modulates the clinical characteristics and outcome of SSc patients. The Spanish Scleroderma Study Group recruited 1037 patients with a mean follow-up of 5.2 ± 6.8 years. Based on the mean ± 1 standard deviation (SD) of age at disease onset (45 ± 15 yr) of the whole series, patients were classified into 3 groups: age ≤ 30 years (early onset), age between 31 and 59 years (standard onset), and age ≥ 60 years (late onset). We compared initial and cumulative manifestations, immunologic features, and death rates. The early-onset group included 195 patients; standard-onset group, 651; and late-onset, 191 patients. The early-onset group had a higher prevalence of esophageal involvement (72% in early-onset compared with 67% in standard-onset and 56% in late-onset; $p = 0.004$), and myositis (11%, 7.2%, and 2.9%, respectively; $p = 0.009$), but a lower prevalence of centromere antibodies (33%, 46%, and 47%, respectively; $p = 0.007$). In contrast, late-onset SSc was characterized by a lower prevalence of digital ulcers (54%, 41%, and 34%, respectively; $p < 0.001$) but higher rates of heart conduction system abnormalities (9%, 13%, and 21%, respectively; $p = 0.004$). Pulmonary hypertension was found in 25% of elderly patients and in 12% of the youngest patients ($p = 0.010$). After correction for the population effects of age and sex, standardized mortality ratio was shown to be higher in younger patients. The results of the present study confirm that age at disease onset is associated with differences in clinical presentation and outcome in SSc patients.

[Late-age onset systemic sclerosis].

PMID: 31586983 | Indexed in PubMed

Systemic sclerosis is a multi-organ connective tissue disease characterised by dysfunction and impaired morphology of the blood vessels with non-specific inflammation and progressive fibrosis. In the majority of cases, the onset is observed between 30-50 years of age; in many cases, however, the diagnosis is established in patients 75 years of age. The course of late-onset systemic sclerosis is markedly different from that in early- onset disease. In late-onset patients, limited systemic sclerosis, pulmonary hypertension, primary heart involvement, and anti-centromere antibodies are more commonly observed. Moreover, the diagnosis of systemic sclerosis in patients > 60 years of age is associated with poor prognosis, higher mortality rates, and an increased risk of neoplasms, as compared to younger patients.

Clinical and laboratory features of patients with scleroderma and silicone implants.

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We reviewed the available clinical and laboratory data from 56 patients with scleroderma and silicone implants from the English medical literature and 19 cases which have not been previously reported. The average age of onset of scleroderma was 43.6 +/- 10 years (range 20-73). Patients had silicone implants for an average of 9 +/- 4 years prior to the development of scleroderma (range 1-32). Most patients had limited scleroderma (41%). Twenty three percent had intermediate scleroderma and 36% had diffuse scleroderma. Clinical findings included: Raynaud's phenomenon in 77%, esophageal dysfunction in 53%, and pulmonary involvement in 47%. Cardiac and renal involvement were uncommon. Antinuclear antibodies by immunofluorescence were found in 83 percent of patients. The immunofluorescence pattern was speckled in 53%, centromere in 31% and nucleolar in 9%. Other antibodies (Scl-70, RNP, SSA/Ro, PM-Scl) were found in only a small proportion of patients. A clinical, serologic and immunogenetic comparison of patients with silicone implants and scleroderma and patients with idiopathic scleroderma is needed to better understand the pathogenesis of this disorder.

[Autoimmune and inflammatory pathologies associated with systemic scleroderma: Clinical, serological and prognostic profiles. Bi-centric retrospective series in the PACA region].

PMID: 37100631 | Indexed in PubMed

Systemic sclerosis (SSc) is a rare auto-immune disease, affecting principally women between 40 and 60 years old. It is caracterised by a cutaneous and visceral fibrosis, an alteration of the microvascular network and the presence of autoantibodies. SSc can be associated with another connectivite tissue disease or to other autoimmune diseases, thus defining the overlap syndrome. The goal of our study is to describe these overlap syndromes. We have analysed the data of a retrospective and bicentrique cohort, from the internal medicine unit of Hôpital Nord in Marseille and from the internal medicine unit of the Hôpital Sainte-Anne in Toulon, of patients followed for a SSc between January 1st, 2019 and December 1st, 2021. We have collected clinical, imunological features, associated auto-immune and inflammatory diseases with its morbidity and mortality. The cohort included 151 patients including 134 limited cutaneous SSc. Fifty-two (34.4%) patients presented at least one associated auto-immune or inflammatory disease. The association of two connectivite tissue diseases including SSc was found in 24 patients (15.9%), a third with Sjögren's syndrome and a third with autoimmune myositis. The principal associated disease to SSc was the autoimmune thyroiditis found in 17 patients (11.3%). The occurrence of complications (hospitalization, long-term oxygene therapy, death) was not significantly different depending on the existence or not of an overlap syndrom. SSc is often associated with other autoimmune diseases. This interrelation between associated pathologies and SSc, modifying sometimes the evolution of SSc, enhances the need of a personalized follow-up.

Exploring the incidence and characteristics of urolithiasis in the central region of Saudi Arabia: Insights from a prominent medical center.

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Urolithiasis is a common and recurrent condition with a rising global incidence. Stones typically develop in the upper urinary tract, primarily the kidneys. Various factors such as age, gender, diet, fluid intake, climate, occupation, genetics, and metabolic diseases influence stone formation. Stones can vary in size and location, causing obstruction, urine stasis, and complications such as infection. The prevalence of urolithiasis in Saudi Arabia has significantly increased in recent decades, and the study aims to determine the current prevalence and composition trends of urolithiasis, guide treatment and prevention strategies, as well as understand predictors of occurrence and recurrence. It is a retrospective cohort study where the data was collected in the time frame of 2015-2021. The study was conducted in the Department of Surgery and the Division of Urology at King Abdulaziz Medical City in Riyadh, Kingdom of Saudi Arabia. The study reveals significant trends in the sociodemographic profile and clinical aspects of urolithiasis patients. With a higher incidence among males (68.5%). Stone compositions predominantly consist of calcium oxalate (67.8%) and uric acid (19.7%), while site distribution shows the left kidney as the most common location (36.5%). Notably, hypertensive patients exhibit a significant association with stone site (P = 0.014). Encouragingly, the majority of patients do not experience reoccurrence (91.6%), and the study demonstrates an increasing recurrence rate with subsequent visits. The relatively shorter hospital stays (55.9% with 1-day stays) indicate efficient management, and this knowledge can aid in optimizing patient care. This study sheds light on the multifaceted nature of urolithiasis by examining

various facets. Low recurrence rate of kidney stones offers positive prospects for effective initial management. The shorter hospital stays, suggest advancements in medical practices, enhancing patient convenience and healthcare resource optimization. Investigating the underlying causes behind the observed stone compositions yield insights into potential preventive strategies. Furthermore, extended studies examining the impact of lifestyle modifications and medical interventions on stone recurrence could contribute to refined treatment protocols. These findings can guide healthcare professionals in optimizing patient care, preventive strategies, and future research endeavors.

Urological Guidelines for Kidney Stones: Overview and Comprehensive Update.

PMID: 38398427 | Indexed in PubMed

Evidence-based guidelines are published by urological organisations for various conditions, including urolithiasis. In this paper, we provide guidance on the management of kidney stone disease (KSD) and compare the American Urological Association (AUA) and European Association of Urologists (EAU) guidelines. We evaluate and appraise the evidence and grade of recommendation provided by the AUA and EAU guidelines on urolithiasis (both surgical and medical management). Both the AUA and EAU guidelines provide guidance on the type of imaging, treatment options, and medical therapies and advice on specific patient groups, such as in paediatrics and pregnancy. While the guidelines are generally aligned and based on evidence, some subtle differences exist in the recommendations, but both are generally unanimous for the majority of the principles of management. We recommend that the guidelines should undergo regular updates based on recently published material, and while these guidelines provide a framework, treatment plans should still be personalised, respecting patient preferences, surgical expertise, and various other individual factors, to offer the best outcome for kidney stone patients.